#1. (a) $\log f(x|\lambda) = \log \frac{\lambda^x e^{-\lambda}}{x!} = x \log \lambda - \lambda - \log x!$

[4] $\theta = \log \lambda \quad \phi = 1$

$$= \frac{x\theta - b(\theta)}{a(\phi)} - c(x, \phi)$$

$b(\theta) = \lambda = e^\theta \quad a(\phi) = \phi \quad c(x, \phi) = -\log x!$

This is in the form of an exponential family.

(b) $b'(\theta) = e^\theta = \mu \qquad \theta = (b')^{-1}(\mu) = \log \mu = g(\mu)$

[3]

(c) $b'(\theta) = e^{\theta} = \mu \qquad b''(\theta) = e^{\theta}$

[3] $V(\mu) = b''(\theta) = b''\left((b')^{-1}(\mu)\right) = b''(\log \mu)$

$= e^{\log \mu} = \mu$

#2 (a) $\log L(\lambda) = \log \prod_{i=1}^{n} p(x_i|\lambda) = \log \prod_{i=1}^{n} \frac{\lambda^{x_i} e^{-\lambda}}{x_i!}$

[2] $= \sum_{i=1}^{n} x_i \log \lambda - n\lambda - \sum_{i=1}^{n} \log x_i!$

(b) $\frac{\partial}{\partial \lambda} \log L(\lambda) = \frac{\sum_{i=1}^{n} x_i}{\lambda} - n$

[2]

$$\frac{\partial^2}{\partial \lambda^2} \log L(\lambda) = -\frac{1}{\lambda^2} \sum_{i=1}^{n} x_i \qquad I(\lambda) = -E\left(-\frac{1}{\lambda^2} \sum_{i=1}^{n} x_i\right)$$

$$= \frac{1}{\lambda^2} n \cdot \lambda = \frac{n}{\lambda}$$

(c) $\frac{\partial}{\partial \lambda} \log L(\lambda) = \frac{\sum_{i=1}^{n} x_i}{\lambda} - n = 0 \Rightarrow \sum_{i=1}^{n} x_i - \lambda n = 0$

[4]

MLE $\hat{\lambda} = \frac{\sum_{i=1}^{n} x_i}{n}$

$$\hat{\lambda} \overset{d}{\simeq} N\left(\lambda, \frac{\lambda}{n}\right)$$

#3. (a) $D = -2\log L(\hat{\theta}) + 2\log L(\text{full})$

[4] $AIC = -2\log L(\hat{\theta}) + 2 \cdot p$

$-2\log L(\hat{\theta}) = AIC - 2 \cdot p = 522.14 - 2 \cdot (70 - 59)$

$= 500.14$

$\log L(\hat{\theta}) = -250.07$

$\log L(\text{full}) = \frac{1}{2}\left(D + 2\log L(\hat{\theta})\right) = \frac{1}{2}(70.665 - 500.14)$

$= -214.738$

(b) Model A (1) AIC is smaller and

[3] (2) Chisq test rejects hypothesis that model B is correct.

(c) $0.02266 \times \sqrt{1.212432} = 0.02495$

[3]

(d) model C, F-test rejects the hypothesis that

[3] model D is correct.

(e) $F = \dfrac{(D_D - D_C)/(df_D - df_C)}{\hat{\phi}} = \dfrac{50.015/3}{1.212432}$

[5]

$= 13.751$

df: 3 & 59

4 (a) $P(\text{test}=1) = \dfrac{e^{\eta_i}}{1+e^{\eta_i}}$

[5]

$\eta_i = -9.879992 + 0.123887 \times 1 + 0.035026 \times 99$

$+ 0.085123 \times 27 + 1.321554 \times 0.25$

$+ 0.023844 \times 25 = -3.063721$

$P(\text{test}=1) = 0.04463$

b) $\frac{O_1}{O_2} = \exp(\eta_1 - \eta_2) = \exp(\beta_{bmi}(27.87 - 36.90))$

[10]

$\hat{\beta}_{bmi} = 0.085123$

95% CI for β_{bmi} : $0.085123 \pm 1.96 \times 0.01811$

$-9.03 \times \hat{\beta}_{bmi} = -0.7687$

95% CI for $-9.03 \times \hat{\beta}_{bmi}$:

-0.7687 ± 0.3205

$\frac{\hat{O}_1}{O_2} = e^{-0.7687} = 0.4636$

95% CI for $\frac{O_1}{O_2}$: $(0.3365, 0.6388)$

So, roughly speaking, all else being equal, the odds of showing evidence of diabetes are between 34 to 64 percent less for a woman with bmi 27.87 compared to a woman wiht bmi 36.90.

#5 (a) see Exam_2018_sol.pdf.
[6]

(b) $p(\theta|x_1 \dots x_n) \propto p(x_1 \dots x_n|\theta)\, p(\theta)$

[6]
$$\propto \prod_{i=1}^{n} \frac{\theta^{x_i} e^{-\theta}}{x_i!}\, \theta^{k-1} e^{-\frac{\theta}{\beta}}$$

$$\propto \theta^{\sum_{i=1}^{n} x_i + k - 1}\, e^{-\left(n+\frac{1}{\beta}\right)\theta}$$

$$\sim \text{Gamma}\left(\left(n+\frac{1}{\beta}\right)^{-1}, \sum_{i=1}^{n} x_i + k\right)$$

$\left(n+\frac{1}{\beta}\right)^{-1} = \beta^*$ $\qquad$ $\sum_{i=1}^{n} x_i + k = k^*$

(c) $E(\theta^{-1}|x) = \int \frac{1}{\theta} \frac{1}{\beta^{*k^*}\Gamma(k^*)} \theta^{k^*-1} e^{-\frac{\theta}{\beta^*}} d\theta$

[6]

$$= \frac{1}{\beta^{*k^*}\Gamma(k^*)} \int \theta^{k^*-2} e^{-\frac{\theta}{\beta^*}} d\theta$$

$$= \frac{1}{\beta^{*k^*}\Gamma(k^*)} \frac{\beta^{*k^*-1}\Gamma(k^*-1)}{1} \int \frac{1}{\beta^{*(k^*-1)}\Gamma(k^*-1)} \square \, d\theta$$

$$= \frac{1}{\beta^*} \frac{1}{(k^*-1)} = \left(n+\frac{1}{\beta}\right) \frac{1}{\sum_{i=1}^{n} x_i + k - 1} = 1$$

$$\hat{\theta} = E(\theta^{-1}|x)^{-1} = \frac{\sum_{i=1}^{n} x_i + k - 1}{n+\frac{1}{\beta}}$$

#6 [7]

1) $X \sim U(0, 2)$

$Y \sim U(0, \log^3)$

2) If $Y < \log^{1+X}$, return X.

Otherwise go back to step 1.

#7 [6]

$\pi(\theta) \propto e^{-\frac{\theta^2}{2}}$

$\pi(\theta) \sim \underline{N(0,1)}$

MH algorithm

i) Start with an initial value $\theta^{(0)}$; set $t=0$.

ii) Given the current value of $\theta^{(t)} = \theta_0$, generate a proposed new value θ_n from a Normal $(\theta_0, 1)$.

iii) Define the acceptance probability A by

$$A = \min\left(1, \frac{\exp\left(\frac{\theta_0^2}{2}\right)}{\exp\left(\frac{\theta_n^2}{2}\right)}\right).$$

iv) With probability A, set $\theta^{(t+1)} = \theta_n$; otherwise set $\theta^{(t+1)} = \theta_0$.

v) Increase t: return to step ii).

#8 (a) $\theta \mid \mu, x \propto \exp\left(-\frac{(\theta-(\mu+1))^2}{(\mu+1)x^2}\right)$

[12] $\sim N\left(\mu+1, \frac{(\mu+1)x^2}{2}\right)$

$\mu \mid \theta, x \Rightarrow$

$$p(\mu=0 \mid \theta, x) = \frac{p(\mu=0, \theta \mid x)}{p(\theta \mid x)} = \frac{C \cdot \exp\left(-\frac{(\theta-1)^2}{x^2}\right)}{p(\theta \mid x)}$$

$$p(\mu=1 \mid \theta, x) = \frac{p(\mu=1, \theta \mid x)}{p(\theta \mid x)} = \frac{C \exp\left(-\frac{(\theta-2)^2}{2x^2}\right)}{p(\theta \mid x)}$$

So, $\mu \mid \theta, x \sim$ Bernoulli $\left(\frac{\exp\left(-\frac{(\theta-2)^2}{2x^2}\right)}{\exp\left(-\frac{(\theta-1)^2}{x^2}\right) + \exp\left(-\frac{(\theta-2)^2}{2x^2}\right)}\right)$

(b) [6]

Start with \theta(0), \mu(0).
Do the following step for n = 0 to M - 1

Given \theta(n), \mu(n),

1. Generate \theta(n+1) from the conditional distribution \theta | \mu(n), x .
2. Generate \mu(n+1) from the conditional distribution \mu | \theta(n+1), x .

Check the convergence, and discard the first B samples as burn-in period. We will use the remaining M-B samples to estimate the mean of \theta/\mu. Specifically,

$$\begin{pmatrix} \hat{\theta} \\ \hat{\mu} \end{pmatrix} = \frac{1}{M-B} \sum_{i=B+1}^{M} \begin{pmatrix} \theta_{(i)} \\ \mu_{(i)} \end{pmatrix}$$